

Response to Referee Report

“Hedged Random Forest”

Journal of Financial Econometrics — JFEC-2025-167

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We thank the referee for the careful reading and constructive suggestions. The report has led to meaningful improvements in the paper. Below we address each point in turn. All section, equation, and table references are to the revised manuscript.

Main Comments

Comment 1

Why not carry out estimation on a second sample using standard OLS with constraints on the weights, which is also a standard approach (Timmermann, 2006)? Ridge would require to tune a parameter, but this could be done by AIC, though ad hoc. Of course, this is a different estimator, but a reader may want to understand the distinction and whether one approach is better than the other.

Response. We have implemented a second-sample benchmark in the spirit of Timmermann [2006]. The approach splits the training data in half: the random forest is trained on the first half, and the tree-level weights are estimated on the second half (which the trees have never seen).

A pure OLS estimator on the second sample is not viable here because the number of trees ($B = 500$) far exceeds the second-sample size ($n/2$), making the system severely underdetermined. Ridge regression with a sum-to-one constraint handles the high-dimensional regime gracefully; we select the penalty parameter λ by AIC using the analytical hat-matrix trace for the equality-constrained problem. We refer to this estimator as *Ridge_{2nd}*.

The empirical results are reported in [TABLE/FIGURE REF]. [RESULTS DISCUSSION PENDING — fill in once final experiment is complete.]

Comment 2

Why not carrying out estimation on the same sample using a ridge regression? Again, helping the reader see the distinction and the merits of one method over the other could be useful.

Response. We have added a same-sample ridge regression benchmark (Ridge). As with the second-sample variant, we impose the sum-to-one constraint and select λ by AIC. Because the constraint is an equality, the standard ridge hat matrix must be corrected; we derive the exact effective degrees of freedom analytically via the economy SVD of the tree-prediction matrix.

The results are reported alongside the other benchmarks in [TABLE/FIGURE REF]. [RESULTS DISCUSSION PENDING — fill in once final experiment is complete.]

Comment 3

Why not focusing on variance minimisation, as for a minimum variance portfolio? Once the weights are derived this way and fixed, the bias, which is a constant, could be removed in a second step, when forecasting.

Response. We have implemented the minimum variance (MV) portfolio as an additional benchmark. The weights are obtained by minimising $\mathbf{w}^\top \hat{\Sigma} \mathbf{w}$ subject to $\mathbf{1}^\top \mathbf{w} = 1$ and the leverage constraint $\|\mathbf{w}\|_1 \leq \kappa$, with $\kappa \in \{1, 2\}$. The systematic bias of the combined predictor, $\hat{b} = \mathbf{w}^\top \bar{\mathbf{r}}$ where $\bar{\mathbf{r}}$ denotes the vector of mean training residuals, is then subtracted from each test prediction. We implement both the sample and the shrunk (Ledoit–Wolf NLS) covariance variants, mirroring the HRF design.

The results are reported in [TABLE/FIGURE REF] and discussed in Comment 4 below.

Comment 4

Individual trees tend to have low bias but high variance, so the variance component may dominate in practice. This justifies the use of the sample mean for the estimator of the bias of the individual trees, i.e. they are small numbers close to zero dominated by the variance. Would the estimator perform the same or perhaps better if we just used a minimum variance portfolio approach and forgot about the bias?

Response. The empirical results for the MV benchmark described in Comment 3 provide a direct answer. Across all datasets and sample sizes, the performance of MV (shrunk, $\kappa = 2$) and HRF (shrunk, $\kappa = 2$) is virtually identical, confirming the referee’s intuition that the bias term in the joint objective contributes little in practice. Individual tree biases are indeed small relative to the variance, so the HRF objective is effectively dominated by the variance component.

This finding is informative in two directions. First, it validates the HRF design: the method works well even though the bias estimation step is noisy, because the variance component is what matters. Second, it provides a useful robustness check for practitioners: a pure variance-minimisation approach with a post-hoc bias correction is equally effective and even simpler to implement.

Comment 5

What is very interesting about negative weights is that they should work better in situations where the trees are correlated. From portfolio optimisation theory we know that the weights will be all positive if the correlations of the trees were small. However, negative weights become more likely as correlations increase. Hence, the authors’ method becomes even more important and relevant when features are correlated, making tree decorrelation more difficult. I think remarking on the properties of portfolio weights (especially in light of Point 4) as a function of correlations would be relevant.

Response. We thank the referee for this insightful observation, which we now make explicit in the paper (Section 4.3).

The connection follows directly from classical results in portfolio theory. When the trees are mutually uncorrelated, the minimum-variance weights are all positive: there is no benefit to short positions because each tree contributes independent variance reduction. As pairwise tree correlations increase, the optimal portfolio can reduce variance further by going short on the most “redundant” trees, using their predictions to cancel noise shared with other trees. The leverage constraint $\|\mathbf{w}\|_1 \leq \kappa$ controls the degree to which this is permitted: $\kappa = 1$ restricts weights to be non-negative (no short-selling), while $\kappa = 2$ allows negative weights.

In the random forest setting this has a natural interpretation. Trees are constructed to be decorrelated via feature subsampling (the `mtry` parameter), but this decorrelation is imperfect when the input features are themselves highly correlated. In such datasets the trees remain substantially correlated, and the gains from $\kappa = 2$ over $\kappa = 1$ are expected to be largest. Conversely, when features are nearly independent, trees are already well-decorrelated, $\kappa = 1$ suffices, and the additional flexibility of $\kappa = 2$ adds little. This provides a characterisation of when HRF with negative weights is most valuable, and it is consistent with the empirical patterns we observe across the benchmark datasets. The corresponding passage has been added to Section 4.3 of the revised manuscript.

Additional Comments

Additional Comment 1

It would be interesting to know whether one can learn something about the methodology and the HRF estimator from the individual characteristics of the datasets (e.g., variable correlations, fat-tailed data etc.). For instance, are there certain data features that systematically relate to larger performance gains?

Response. [RESPONSE PENDING]

Additional Comment 2

Many readers will naturally associate the leverage constraint with a Lasso-type penalty and therefore expect a sparse solution, i.e. that only a small number of trees receive nonzero weights. However, the additional portfolio equality constraint (i.e., the requirement that

the weights sum to one) changes the geometry of the optimisation problem, and often leads instead to a dense weighting scheme. It might be helpful to include a short remark about this.

Response. We have added Remark 3.4 to Section 3 of the revised manuscript.

The key distinction is geometric. An L1 penalty alone (without any equality constraint) encourages sparsity because the corners of the L1 ball align with coordinate axes: the penalised objective is minimised at a vertex, where all but a few weights are zero. The sum-to-one equality constraint $\mathbf{1}^\top \mathbf{w} = 1$, however, restricts the feasible region to a hyperplane that generically passes through the interior of the L1 ball rather than through its sparse vertices. The constrained solution therefore lies on the hyperplane, which is a full-dimensional affine subspace; there is no geometric pressure towards the coordinate-axis corners, and the solution is typically dense.

Intuitively, the difference between Lasso and HRF is analogous to the difference between asset selection (Lasso finds a sparse subset of trees with non-zero weight) and portfolio construction (HRF assigns a weight to every tree, possibly including short positions, to minimise prediction error).

Additional Comment 3

In section 3, the authors define their specific implementation of the estimator: shrunk covariance matrix and a leverage constraint equal to 2. Later in the paper, they compare results with a leverage constraint equal to one and also compare to the estimator that uses the standard sample covariance matrix. A number of different intuitive notations are used to define these. It would be ideal if this notation or some similar version could be used everywhere. Using shrinkage/sample and κ (the leverage constraint) equal to 1 or 2, everywhere is a possible solution.

Response. We thank the referee for this suggestion. We have unified the notation throughout the paper along the lines proposed. Each variant of the estimator is now identified by two labels: the covariance estimator used (“shrinkage” for the Ledoit–Wolf NLS estimator, “sample” for the plain sample covariance) and the leverage parameter $\kappa \in \{1, 2\}$. For example, the main HRF specification discussed in Section 3 is now consistently referred to as HRF (shrinkage, $\kappa = 2$), and this labelling is used uniformly in the text, all tables, and the appendix, including the notation tables in Section 4.2 and the appendix.

References

Allan Timmermann. Forecast combinations. In G. Elliott, C. W. J. Granger, and A. Timmermann, editors, *Handbook of Economic Forecasting*, pages 135–196. North Holland, Amsterdam, 2006.